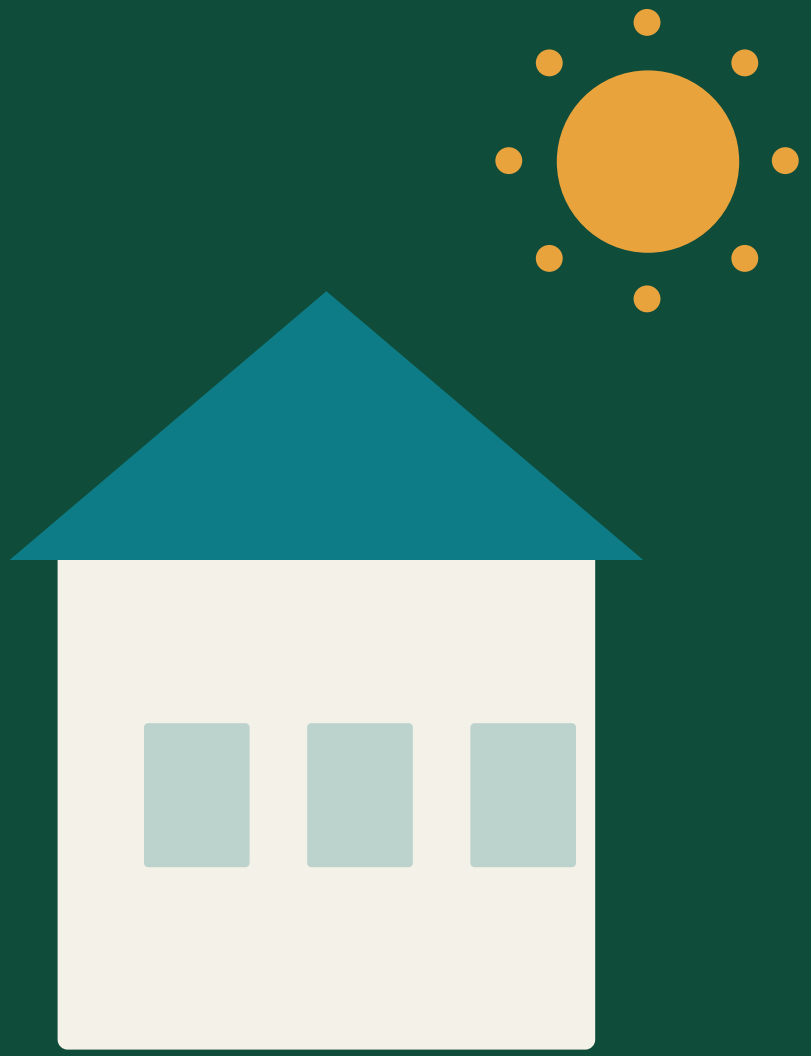


Cooler roofs · lower bills · clearer quotes

HACKATHON PROJECT

# Cool Roofing Intelligence

Cost, comfort and energy-savings predictions before the first quote

Learned from 2,500 real installations

5 prediction tasks · 6 models · explainable · tested

THE PROBLEM

# A roofing quote still rests on four open questions

## How much will it cost?

Installation cost depends on size, method, quality and roof.

## How much cooler will the roof get?

Cooling varies by method and quality, not just by price.

## How many kWh will we save?

Savings depend on current usage and AC load, and can be near zero.

## Will they rate us highly and refer us?

Satisfaction and referrals are driven by experience after the work.

**The opportunity**

2,500 past installations record cost, roof temperature and electricity use before and after, plus the customer's rating and referral. That is enough to learn from, if the data can be trusted and the answers are explained honestly.

Source: cool_roofing_problem.md; 2,500 historical installation records

THE SOLUTION

# One system that predicts, compares, explains and proposes

**CORE: implemented and tested**

- 5 prediction tasks → 6 models
- Data audit and cleaning pipeline
- Hold-out evaluation vs baselines
- Roofing-option comparison engine
- Per-customer explanations
- Payback calculator (your tariff)
- PDF customer proposals
- React app + FastAPI service

**OPTIONAL: implemented**

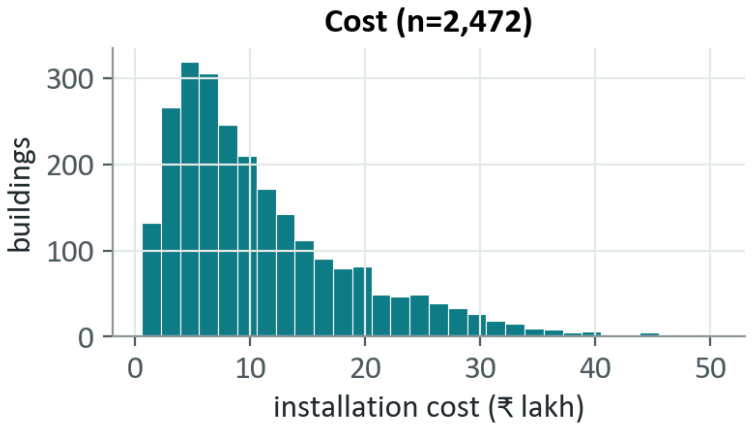
- AI sales assistant: built-in free mode, plus Gemini (live-tested)
- Claude integration built, not live-tested (no credit)
- Assistant only calls validated tools; it cannot invent predictions

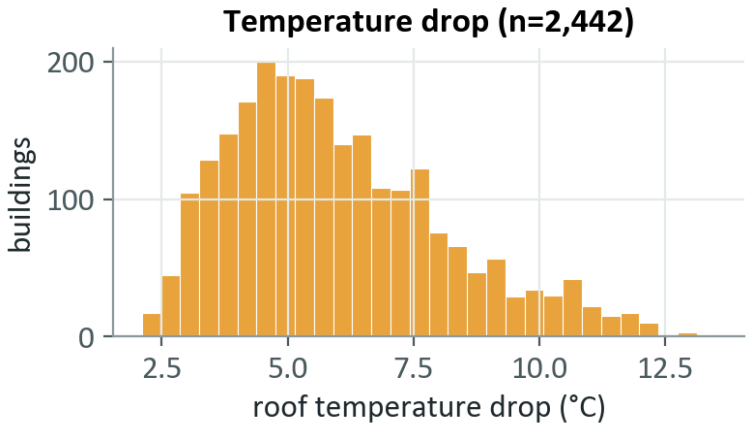
**FUTURE: not built**

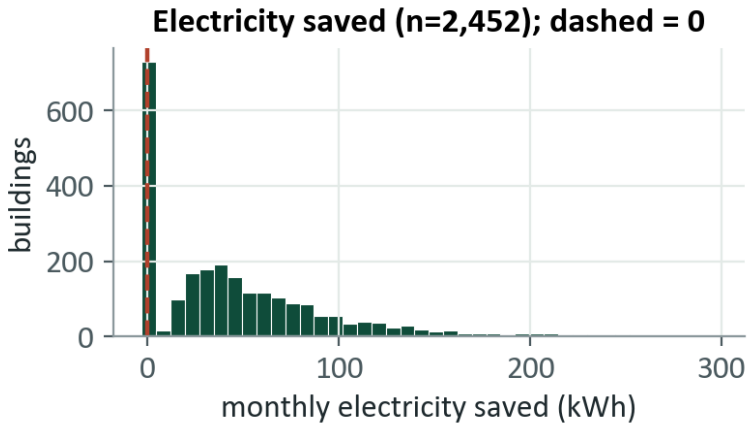
- Digital-twin simulator
- Carbon-savings tracker
- Geo-climate intelligence
- Validation on newly measured installations

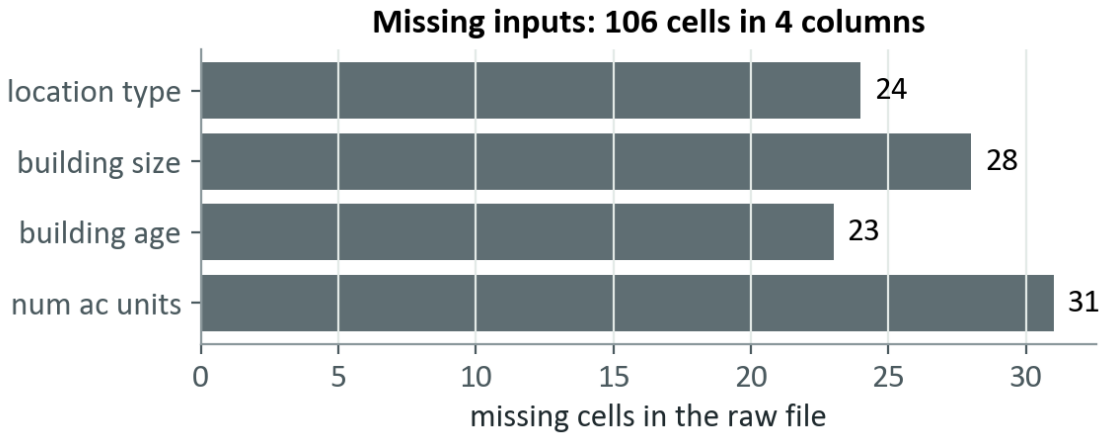
Status taken from the repository: implemented and tested vs. optional vs. future

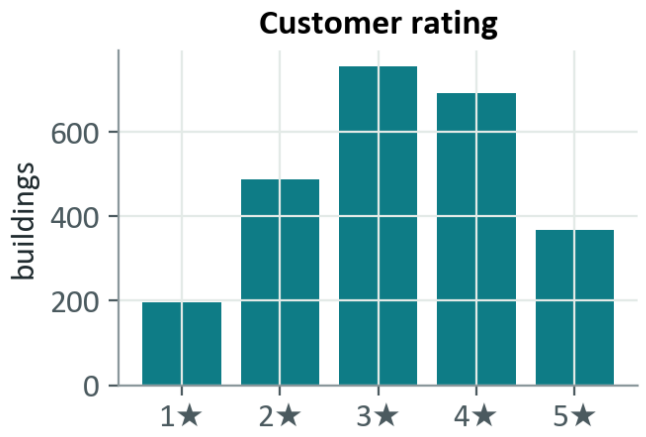
# 2,500 installations, 15 columns, three measurable outcomes

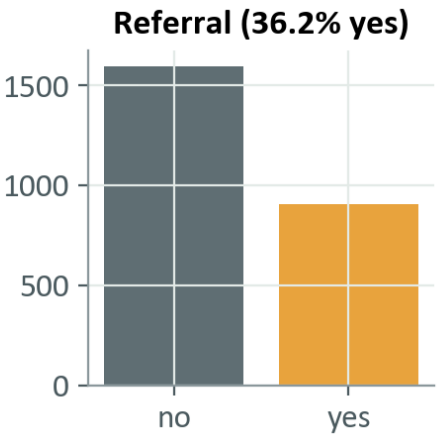


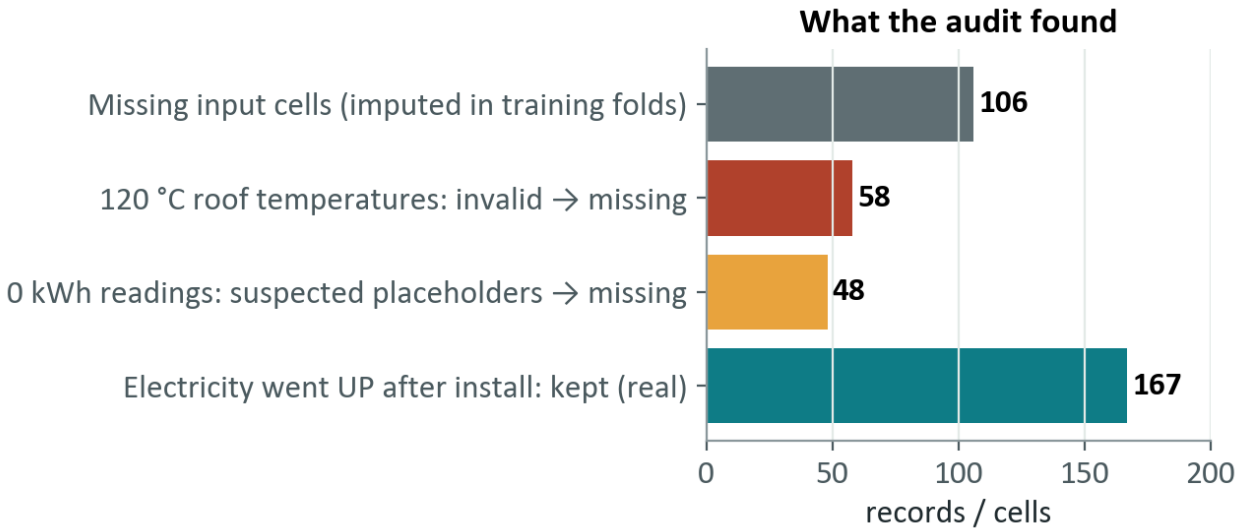



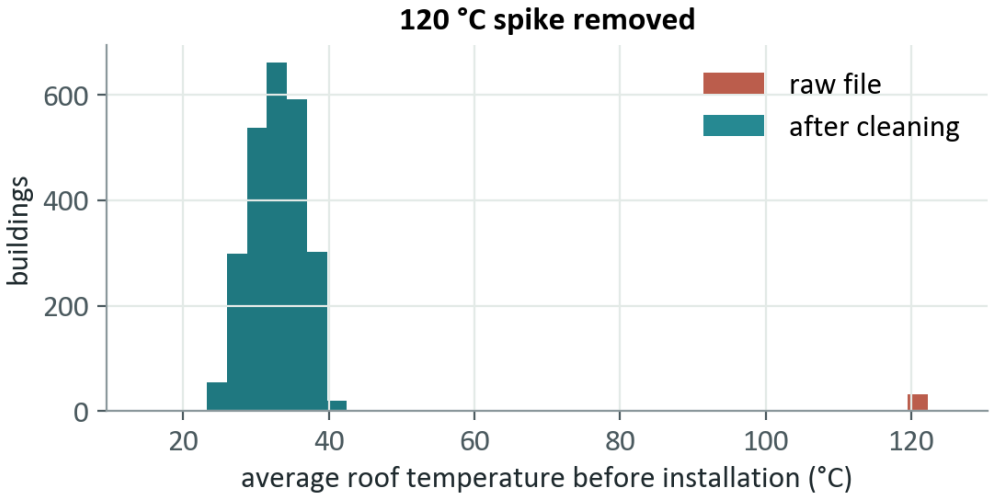

Records: 2,500 · Inputs: location, size, age, existing roof, method, quality, start temperature, AC units, current kWh · Outcomes: cost, temperature, electricity, rating, referral

Source: dataset/cool_roofing_data.csv (read-only); charts computed from reports/data_profile.json after cleaning

DATA QUALITY

# The audit found two silent placeholders that would have corrupted the models




## Confirmed invalid

120 °C roof temperature: 58 records (32 before, 26 after)
Set to missing, never capped or guessed

## Inferred, high confidence

0 kWh: 48 readings. No 1–3 kWh values exist; rows are physically inconsistent
Effect: savings MAE 4.45 → 3.62 kWh

## Kept on purpose

167 buildings used MORE electricity after install: real adverse outcomes
0 records deleted; exclusions are per task only

Source: reports/data_quality_report.md, reports/cleaning_audit.csv (record-level audit trail); raw file never modified

# Only what a rep knows at the table can be a model input

**Temperature drop (°C)**

= roof temperature before − roof temperature after

**Electricity saved (kWh / month)**

= monthly kWh before − monthly kWh after

## Evaluation protocol

**2,500 buildings**

**2,000 train + validation**
**5-fold CV: tuning and selection**

**500**
**hold-out**

- Preprocessing fitted on training folds only
- Hold-out scored once; baselines declared first
- Shuffled-label and ID-only controls sit at chance

## ALLOWED at quote time (9 inputs)

| | | |
|---|---|---|
| Location type | Building size | Building age |
| Existing roof | Proposed method | Proposed quality |
| Start roof temp. | AC units | Current kWh |

## FORBIDDEN as inputs

| | | |
|---|---|---|
| Realised cost | Temp. after install | kWh after install |
| Customer rating | Referral flag | Building ID |

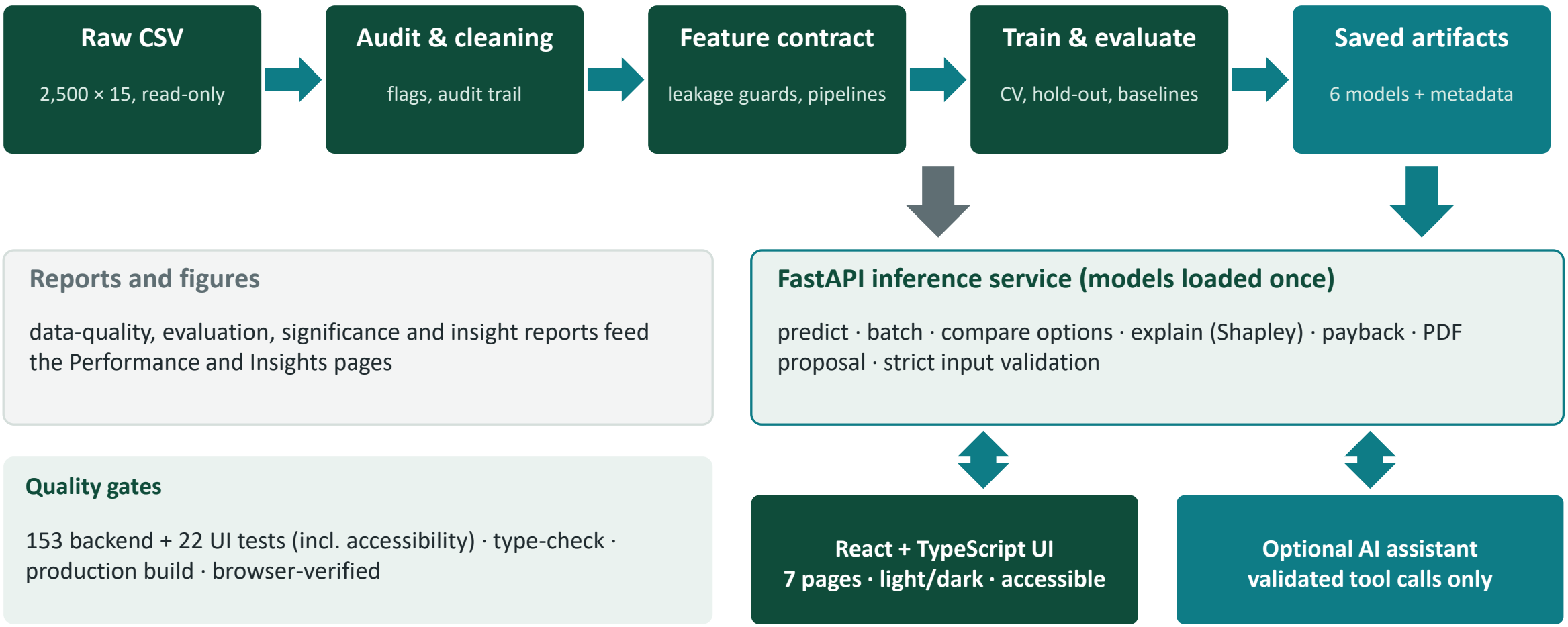
**Why it matters:** referral rate is 61% for 4★+ customers vs 9% for 1–2★, so rating would leak referral.

Rejected at the API too: unknown fields return HTTP 422.

Source: docs/feature_contract.md; tests/test_features_leakage.py, tests/test_model_review.py

SYSTEM ARCHITECTURE

# From raw CSV to a tested, explainable web product




pandas · scikit-learn · LightGBM · FastAPI · React 19 + TypeScript + Vite · Tailwind · Recharts · ReportLab

Source: docs/architecture.md and the implemented code (src/coolroof, frontend/)

MODELS

# Five tasks, six models: simple where the data is simple

| # | Task | Prediction | Selected model | Type | Baseline | Train+val / test |
|---|---|---|---|---|---|---|
| **A** | Installation cost | Installation cost (₹) | Log-linear ridge (multiplicative) | Regression | Training mean | 1,979 / 493 |
| **B** | Roof temperature drop | Roof temperature drop (°C) | LightGBM: drop-fraction × starting temperature | Regression | Training mean | 1,956 / 486 |
| **C** | Monthly electricity saved | Electricity saved (kWh/mo) | LightGBM: saved-fraction × current usage | Regression | Training mean | 1,958 / 494 |
| **D1** | Customer rating | Customer rating (1–5) | LightGBM | Regression | Training median | 2,000 / 500 |
| **D2** | High rating | High rating (≥ 4★) | LightGBM | Classification | Majority class | 2,000 / 500 |
| **E** | Customer referral | Customer referral | Logistic regression | Classification | Majority and all-positive | 2,000 / 500 |

**Regression** predicts a number (₹, °C, kWh, stars). **Classification** predicts a yes/no with a probability.
The high-rating threshold of 4 stars is our assumption: the organisers gave no definition (configurable).

Source: models/model_metadata.json. Customer rating needs two models (numeric rating + high/not-high classifier)

EVALUATION METHODOLOGY

# Judged on 500 buildings the models never saw

## MAE

**Mean absolute error**

Average size of the miss, in the target's own unit

## RMSE

**Root mean squared error**

Like MAE, but big misses count more

## Accuracy

**Share correct**

Misleading when one class dominates

## F1

**Precision × recall balance**

Matters for the rarer class, e.g. referrals

Precision = flagged who truly are · Recall = true cases we catch · AUC = ranking skill (0.5 = chance)

**Same split everywhere:** models and their baselines are scored on the identical hold-out rows, so comparisons are direct.

## How we kept it honest

1 500-building hold-out fixed before modelling, scored once

2 Tuning and selection by 5-fold CV on the rest

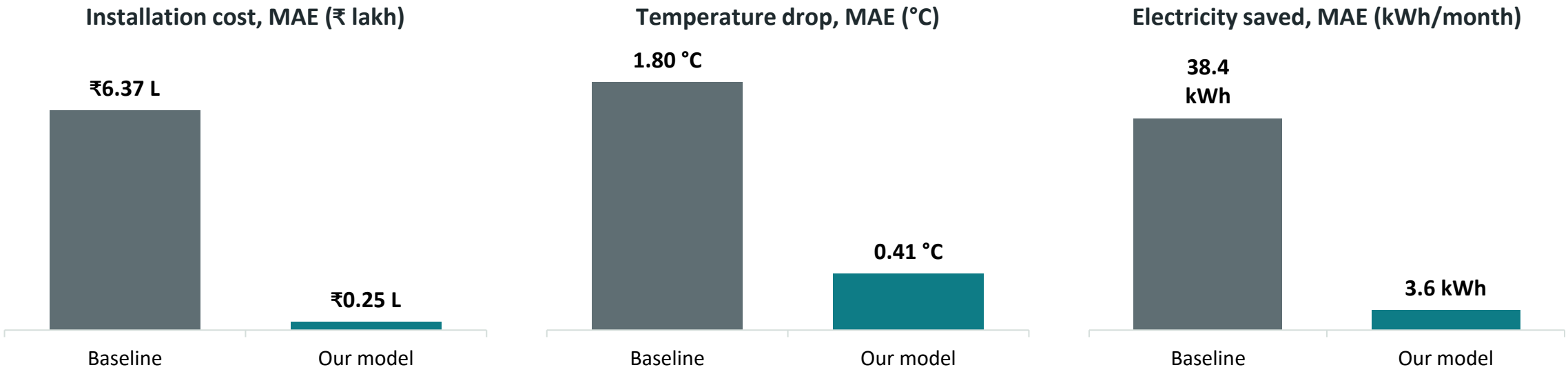
3 Baselines declared first: training mean (median for rating); majority class and 'everyone positive' for classifiers

4 'Better' only if the paired-bootstrap 95% CI excludes 0

5 Independent review recomputed every metric and replicated the selection on never-explored rows

Source: docs/model_evaluation.md and docs/model_review.md; paired bootstrap, 2,000 resamples, 95% intervals

# Regression models cut error 4–25× versus a mean baseline




| Task | Selected model | MAE | RMSE | $R^2$ | Baseline MAE |
|---|---|---|---|---|---|
| **Cost** | Log-linear ridge | ₹25,299 | ₹38,515 | 0.998 | ₹637,024 |
| **Temperature drop** | LightGBM (ratio target) | 0.41 °C | 0.51 °C | 0.948 | 1.80 °C |
| **Electricity saved** | LightGBM (ratio target) | 3.62 kWh | 5.87 kWh | 0.987 | 38.4 kWh |

## What this shows

MAE improvement: 95% CI above 0 for all three
80% ranges cover 78% / 79% / 76% (slightly narrow)
Accuracy comes from legitimate inputs: no size → cost $R^2$ 0.22

Source: models/model_metadata.json (hold-out n = 493 / 486 / 494); each chart has its own unit and scale

# Ratings and referrals are only weakly predictable, and we show it

## Customer rating and high rating

**Rating MAE**

**0.77 ★**

baseline 0.92 · better

**High-rating accuracy**

**0.688**

majority 0.576 · better

**High-rating F1**

**0.614**

all-high 0.596 · n.s.

**High-rating AUC**

**0.76**

95% CI 0.71–0.80 · above chance

**High rating means 4 stars or more: our assumption (no official definition).**

## Customer referral (best-F1 cut-off 25%)

| | Predicted: no | Predicted: yes |
|---|---|---|
| **Actual: no** | 95 | 218 |
| **Actual: yes** | 26 | 161 |

**F1** 0.569 vs 0.544 (everyone refers): not distinguishable

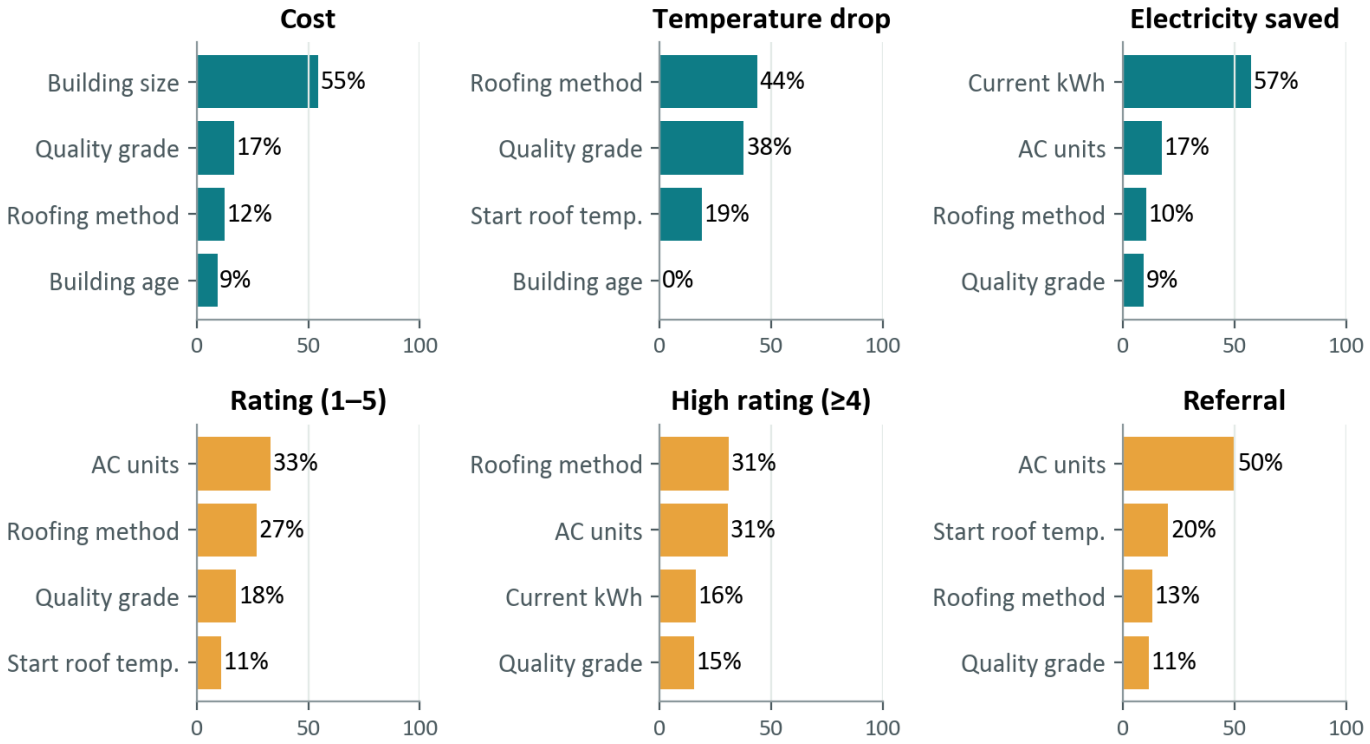
**Accuracy** 0.512 vs 0.626 (majority class): significantly worse

**Recall** 0.86 · precision 0.42 · AUC 0.66 (CI 0.61–0.70)

**Accuracy and F1 pull in opposite directions for referrals. No cut-off beats the trivial classifiers on both, so the app shows both and uses referral probability only to rank follow-ups.**

Source: models/model_metadata.json and reports/significance.json (hold-out n = 500); trivial baselines declared before scoring

# Size drives cost; method and quality drive cooling and satisfaction




## Model view: methods on concrete roofs

| Method | Cost | Cooling | kWh | P(high) |
|---|---|---|---|---|
| **Basic coating** | ₹8.9 L | 4.6 °C | 41 | 35% |
| **Sheet** | ₹9.8 L | 6.0 °C | 51 | 47% |
| **Sheet + coating** | ₹12.5 L | 7.8 °C | 62 | 55% |
| **Tiles** | ₹10.7 L | 5.7 °C | 49 | 42% |
| **Tiles + coating** | ₹13.4 L | 8.2 °C | 65 | 58% |

### Careful reading

Coated options lead on cooling, savings and satisfaction; none wins everywhere

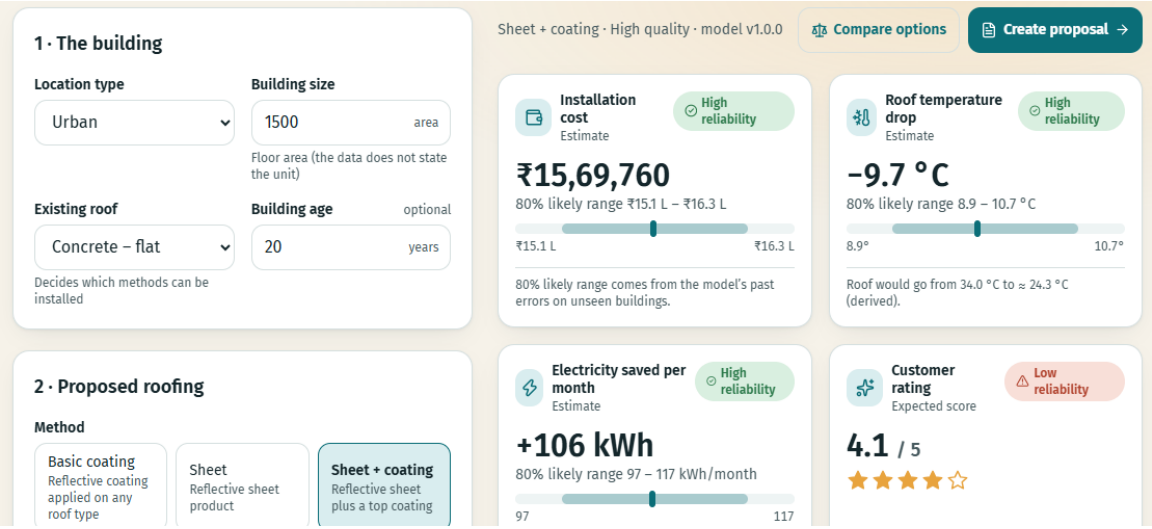
Associations, not causation

Electricity savings alone repay slowly: the case rests on comfort

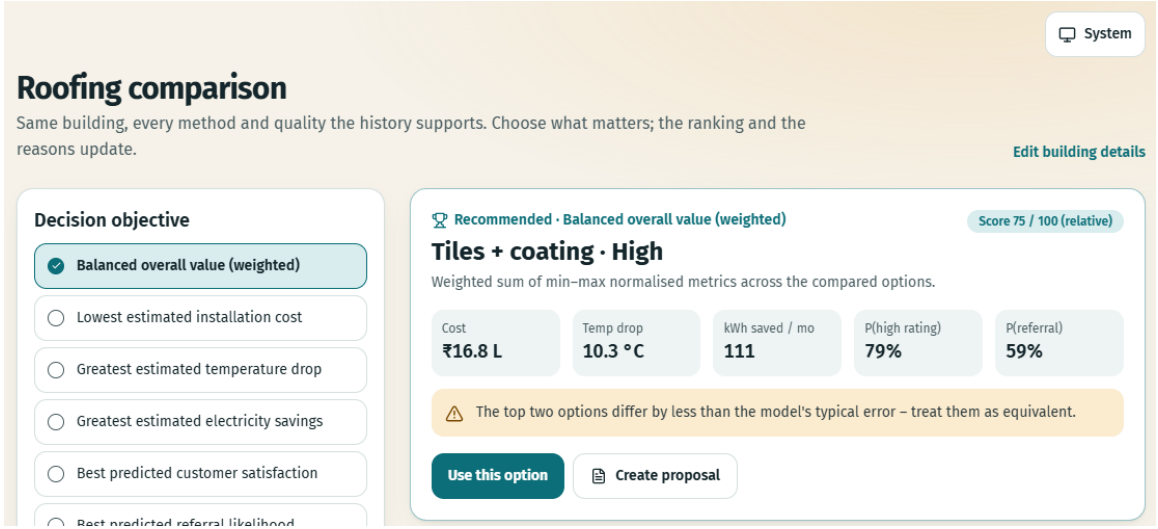
Source: reports/feature_importance.csv (hold-out permutation importance), reports/insights.json (model counterfactuals, concrete roofs, n = 1,323). Associations, not causal effects.

DEMONSTRATION

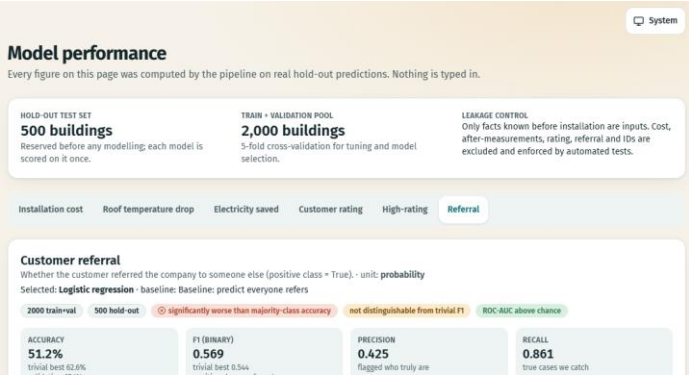
# The product is built and running: predict, compare, explain, propose



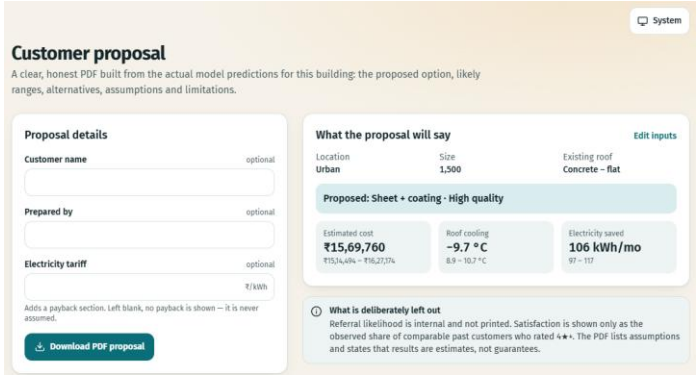

**Customer prediction:** cost, cooling, kWh, rating and referral, each with a likely range and a reliability flag



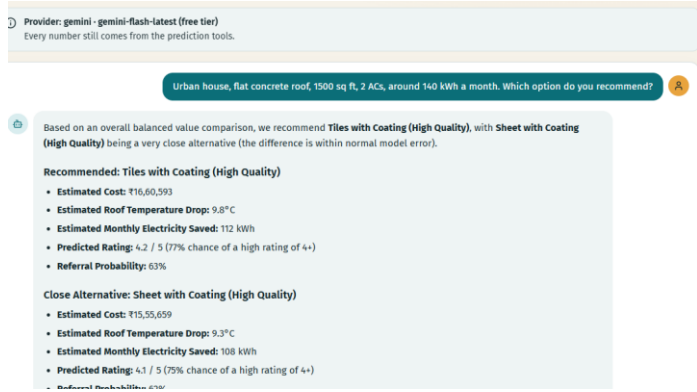

**Roofing comparison:** every valid method × quality, ranked for your objective, with the basis shown




**Model performance:** baselines and intervals




**PDF proposal:** from real predictions




**AI assistant:** Gemini live-tested

Source: screenshots of the running local application (React + FastAPI). Assistant: Gemini live-tested; Claude not live-tested.

# Useful today as decision support; honest about what it cannot do

## Business value (demonstrated)

- Preliminary estimates in well under a second
- Compare 15 method × quality options per building
- Every prediction explained and assumptions visible
- Proposal PDFs from real model outputs
- Payback only from the customer's own tariff

## Limitations (stated openly)

- Estimates, not guarantees
- 0 kWh placeholder treatment is an inference
- Rating and referral only weakly predictable
- High-rating threshold (4★) is our assumption
- Associations, not causation; no time or location fields
- Claude not live-tested (Gemini is)

## Roadmap (future, not built)

- Validate on newly measured installations
- Digital-twin simulator
- Carbon-savings tracker
- Geo-climate intelligence

Source: docs/business_recommendations.md, docs/assumptions_and_decisions.md, docs/model_review.md. Roadmap items are NOT implemented.

CONCLUSION

# From 2,500 past roofs to decisions that are explained and honest

## What we built

- Audit + cleaning pipeline
- 6 models, 5 tasks
- Compare · explain · propose
- 175 automated tests

## What the evidence supports

- Cost MAE ₹25,299
- Cooling MAE 0.41 °C
- Savings MAE 3.62 kWh
- Rating and referral: weak signal

## Strongest capabilities

- Leakage-free, hold-out-verified
- Honest baselines and CIs
- Explanations per customer
- Working end-to-end app

## Next step

- Validate on newly measured installations
- Then: digital twin, carbon tracker, geo-climate

**Historical roofing data, turned into estimates a sales rep can trust, with the reasons and the limits attached.**

## Thank you · Questions?